MILITARY SPECIFICATION SHEET

PASSIVE INTERPHONE CONTROL PANEL

C-2648B/AIC-14

Inactive for new design after 10 August 1984
For new design use MIL-I-22352/17(AS) or /18(AS).

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope

This specification sheet provides performance and physical characteristics for the passive Interphone Control panel C-2648B/AIC-14, hereinafter referred to as C-2648B when used for Receiver Selector operation.

1.2 Military Part Number. The military part number shall be M22352/16-1

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item definition. The passive Interphone Control panel C-2648B shall provide:

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(1) Interconnection with radio receivers. Any combination of one through six receivers may be selected at the operators option.

3.1.1 Item diagram. Passive Control Panel C-2648B shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The passive Interphone Control panel input and output signals will normally display the characteristics presented in Table I.

TABLE I. Interface signal list.

Signal Name	Characteristics
Inputs:	
Receiver No. 1 (RCVR NO. 1)	Voice range audio (300 to 700 Hz) signal ranging in amplitude from 0 to 15 Vac.
Receiver No. 2 through Interphone No. 6 (RCVR NO. X)	Same as Receiver No. 1
Panel Lights (LIGHTS)	0 to ± 28Vdc or 0 to 6V, 400 Hz
Outputs:	
Selected Receiver (SEL RCVR)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Emergency Receiver No. 1, No. 2, and No. 3 (EMER RCVR NO. X)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 15 Vac. No dc component normally present.
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal Return. See 3.4.4.3 of MIL-I-22352(AS).

3.1.3 Controls. The passive Interphone Control panel shall provide the controls listed in Table II.

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TABLE II. C-2648B controls list.

Title	Function
RCVR SEL	The Interphone Control panel shall provide a series of six two position center-off toggle switches as defined herein. The switches shall provide the operator at each Interphone station with the capability to select the following functions: SELECTED - In the up position the operator will receive the interphone audio of the Receiver associated with the switch actuated. OFF - In the down position the system outputs are disabled.

3.1.4 Indicators. The passive Interphone Control panel shall provide two front panel lamps to enhance night vision.

3.2 Characteristics

3.2.1 Performance characteristics

3.2.1.1 Input characteristics

3.2.1.1.1 Receiver No. 1 through Receiver No. 6 (RCVR NO. X). The RCVR NO. 1 through RCVR NO. 6 input characteristics shall be as follows:

3.2.1.1.1.1 Signal Frequency. The RCVR NO. X inputs shall accept drive signals which range from 300 to 6,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The NCVR NO. X inputs shall accept drive signals which range in amplitude from 0 to 20 Vac.

3.2.1.1.1.3 Input Impedance. The RCVR NO. X inputs shall each present a 62,000 ohm ±10 percent impedance to the incoming signals.

3.2.1.1.1.4 Isolation. The isolation between RCVR NO. X signals shall be at least 20 dB for signal ranging from 300 to 6,000 Hz.

3.2.1.1.2 Panel Lights (LIGHTS). The C-2648B shall be capable of accepting power ranging up to +28 Vdc or 6 Vac at 400 Hz to illuminate the two front panel lamps. The unit shall be capable of operating with either ac or dc lighting power source as permitted in MIL-C-6781. When ac power is used for panel lighting self susceptibility to the 400 Hz power source shall induce not more than 5 millivolts across any properly terminated output with the input circuits shorted. The panel lamps used shall be selected from Table III.

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TABLE III. Front panel bulbs.

Nomenclature	Power
MS25237A-327	+28 Vdc
MS25237A-328	6 Vac

3.2.1.2 Output Characteristics. The C-2648B shall have the following output characteristics:

3.2.1.2.1 Selected Receiver (SEL RCVR). The SEL RCVR output shall be the sum of operator selected RCVR NO. 1 through RCVR NO. 6 inputs. Selection is made by setting the desired CRCUR SEL switches to the SELECTED (up). position.

3.2.1.2.2 Emergency Receiver No. 1 through Emergency Receiver No. 3 (EMER RCVR No. X). The C-2648B, when configured for Radio Receiver Selector operation, shall provide emergency operation in which the receiver inputs associated with the three leftmost RCVR switches are available as outputs on the connector as shown in Figure 1.

3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The C-2648B passive Interphone Control panel shall consist of a unit having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The C-2648B shall weigh no more than 1.0 pounds.

3.3 DESIGN AND CONSTRUCTION

The design and construction of the C-2648B passive Interphone Control panel shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

3.3.1 Plastic lighting plates. Plastic lighting plates shall meet the requirements of MIL-P-7788 and Figure 2 of this specification. Markings shown on Figure 2 shall conform with the appropriate application requirements of MS25493. A listing of lighting plates for the C-2648B is contained within Supplement 1 to this specification. If the appropriate configuration drawings are not specified in the contract, this information will be supplied by the acquiring activity upon application by the contractor.

3.3.2 Front panel lighting. The front panel lamps shall be installed using mounting hardware having red filters in accordance with MIL-L-25467 to enhance night vision. The panel lamps shall be mounted in accordance with the outline drawing Figure 2. The lamp drive circuitry shall be capable of accommodating power hanging up to 28 Vdc or 6 V, 400 Hz ac.

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3.3.3 Electrical Connections. A shorting plug designated P601 shall be provided internal to the C-2648B to facilitate use of the C-2648B as a Radio Selector Panel or as an ICS Selector Panel. Plug, P601, shall be installed for use of the C-2648B as an ICS Selector Panel (Refer to MIL-I-22352/15(AS).

4. QUALITY ASSURANCE PROVISIONS
See MIL-I-22352(AS).

4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
Navy - AS

Project No. 5831-N042-16

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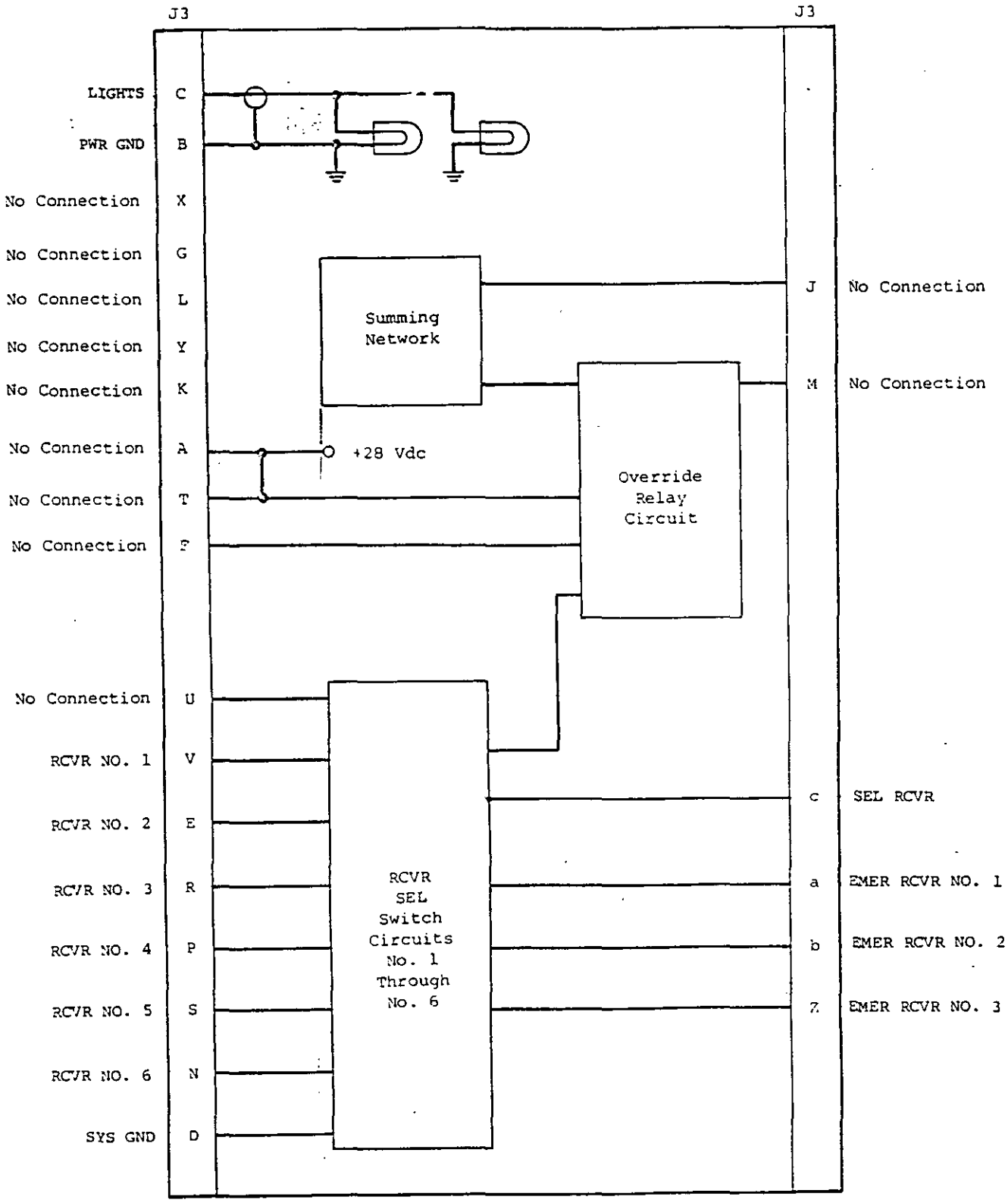
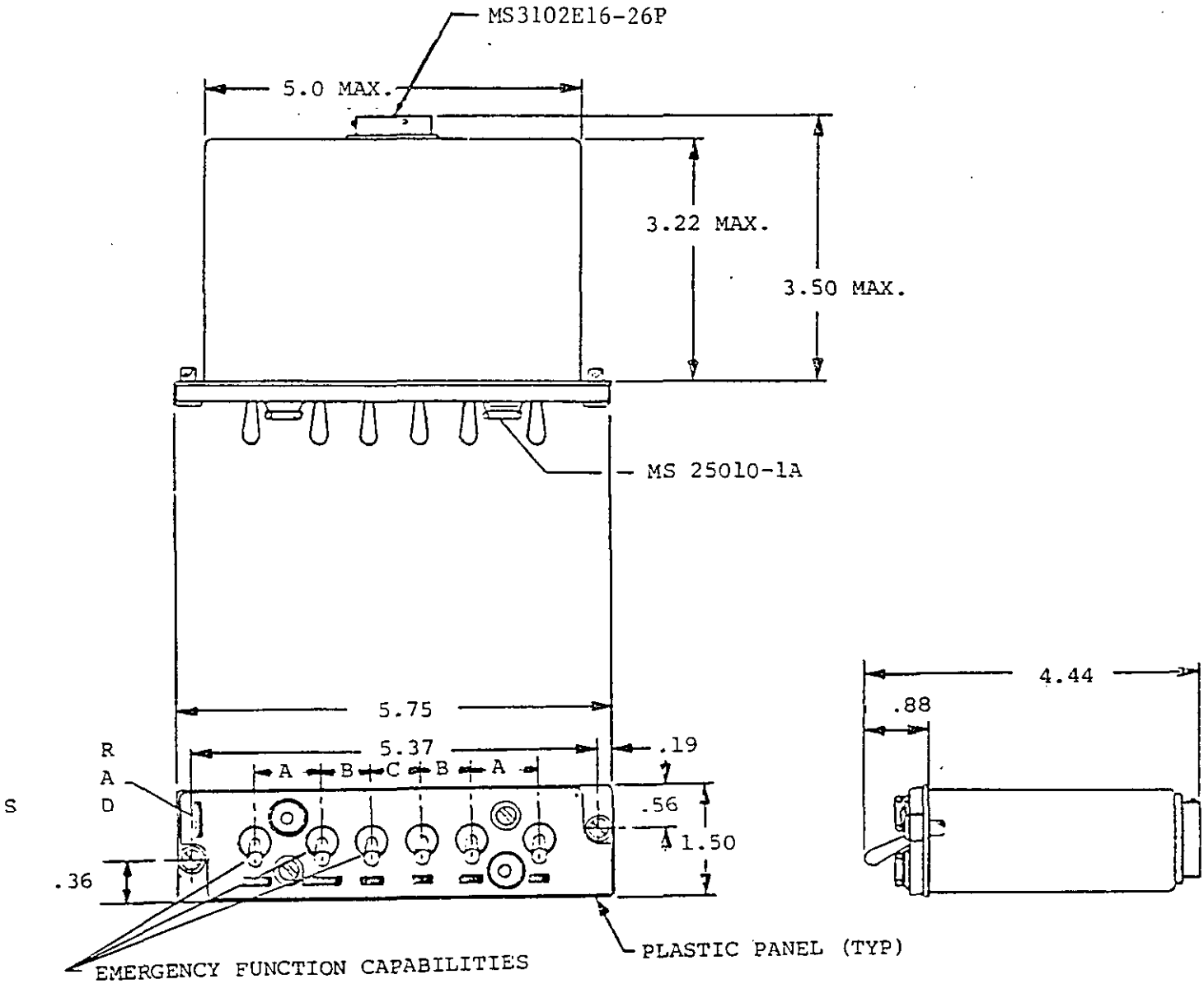


FIGURE 1. C-2648B (Receiver Selector) functional diagram.

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NOTE:

1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. C-2648B outline drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/16(AS)	PASSIVE INTERPHONE CONTROL PANEL C-2648B/AIC-14

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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